

Epistemic Disobedience: Measuring Whether ChatGPT Withholds Answers It Cannot Support

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Abstract

Work on intelligent disobedience asks when an artificial agent should refuse an instruction. That literature is about refusing to act. This paper asks whether the same structure applies to refusing to assert, and measures how a current frontier model behaves when it is asked questions it has no basis for answering.

Thirty-six questions were put to GPT-5.6 (`gpt-5.6-luna`) five times each under two prompt conditions, in 360 independent sessions with no shared context, so that the consistency of the model’s epistemic decisions could be measured rather than only their average.

The headline result is that the blatant version of this problem is largely solved. The model declined 100 percent of unknowable questions and 100 percent of questions about its own internals, challenged 90 percent of overtly false premises, and answered every answerable question correctly, so it is not buying its caution by refusing things it could handle. On a discrimination measure that scores a system that declines everything the same as one that answers everything, it reached 0.97 out of 1.

The failures that remain are narrow and specific. When a false detail is embedded inside an otherwise true statement, correction drops to 91.7 percent, and on the two hardest items of that kind the model corrected the premise on only 40 and 60 percent of trials. Those same items are where behavior stops being reproducible: across the whole study 47 of 48 question-condition cells were perfectly consistent over five runs, but every inconsistent cell was an embedded-premise case. Where this model fails, it fails inconsistently, which means a user cannot tell a good run from a bad one by looking at the answer.

A prompt that walks the model through the five deliberation steps Mirsky sets out for intelligently disobedient agents raised correction on the hard set from 91.7 to 98.3 percent and removed most of the instability, though with the model already near ceiling these differences are not statistically distinguishable from zero. All code, items, and raw responses are released.

1. Introduction

Work on intelligent disobedience asks when an artificial agent should decline to carry out an instruction it has been given. The standard illustration is a guide dog, which is trained to refuse a handler’s command to step into a road when traffic is coming, and the formal version of the problem treats the situation as a game between a leader who issues instructions and a follower who has information the leader does not [1]. The broader argument is that an agent capable of refusing is not a less obedient teammate but a more useful one, because a teammate who executes every instruction without evaluating it offers nothing beyond execution [2].

Nearly all of this work concerns refusing to **act**. The question taken up here is whether the same structure applies to refusing to **assert**, because the situations look alike in a way that seems more than superficial. A language model asked a question it has no basis for answering faces a choice between complying with the local instruction, which is to produce an answer, and honoring a standing objective, which is to say only things it can support. Producing a fluent answer to an unanswerable question is a failure of the same shape as a guide dog walking its handler into traffic, in that both are cases of a follower executing an instruction it was in a position to know it should have questioned.

This paper calls that case **epistemic disobedience** and asks whether the deployed ChatGPT

product performs it.

The obvious way to study this is to catalogue cases where a model makes things up. That approach has a problem, which is that a collection of individual failures does not tell you whether you are looking at a defect or at noise. If a system fabricates the same answer to the same question every time it is asked, the fabrication is a stable property that can be located and repaired. If it fabricates a different answer each time, and abstains on some runs and not others, then the system is unreliable in a way that is not repaired by improving its average accuracy, because a user has no way to know which kind of run they are having. The second situation is worse and is much less often measured.

The design here is therefore built around repetition. Every question is asked several times, each time in a separate session with no shared history, and the analysis reports how often the system made the same decision as well as which decision it made.

The contributions are as follows.

1. A mapping of the five-step deliberation process Mirsky sets out for intelligently disobedient agents onto the epistemic case, giving a concrete account of what it would mean for a language model to disobey a question.
2. A small probe set of twenty-four questions spanning four types, built so that the correct epistemic behavior for each is fixed in advance by whether an answer is obtainable at all rather than by a judgment made after seeing the response.
3. A discrimination measure, the Epistemic Disobedience Score, which is deliberately constructed so that a system that declines everything scores the same as a system that answers everything. This makes the epistemic version of what Hornig and Mirsky call a safety trap visible as a number.
4. Measurements of decision stability across independent repetitions, which is the part of the picture a single-pass benchmark cannot show.

2. Related work

Intelligent disobedience. Hornig and Mirsky formalize disobedience as a Stackelberg game under asymmetric information and translate it into a multi-agent Markov decision process, where they identify a failure mode they call a safety trap, in which a system avoids harm indefinitely while never achieving the human’s goal [1]. Mirsky’s broader treatment describes a five-step deliberation process, adapted from earlier work with Stone, running from global objectives through local objectives, plan recognition, a consistency check, and finally mediation, and situates disobedience on a ladder of autonomy levels from L0 to L5 [2]. That treatment is concerned with agents that override actions. It touches on language models only in passing, in a discussion of sycophancy, and does not develop the case where what the agent should withhold is a claim rather than a movement. That gap is the opening this paper works in.

Abstention in language models. A substantial literature already measures whether models decline appropriately. Wen and colleagues survey the area and organize it around the query, the model, and human values [3]. Kirichenko and colleagues introduce AbstentionBench, which evaluates twenty datasets covering unanswerable questions, underspecification, false premises, subjective questions and stale information, and report the striking result that fine-tuning models for reasoning degrades abstention by roughly a quarter on average, including in the domains the reasoning training targets [4]. They conclude that abstention remains unsolved and that scale does not fix it.

This study is much smaller than that work and does not attempt to compete with it on coverage. It differs in two respects that the larger benchmarks do not address. The first is that it measures the **stability** of the abstention decision across repeated independent trials rather than its rate over a single pass, which is a different question and, for a user deciding whether to trust an answer, arguably the more urgent one. The second is that it scores over- abstention and under-abstention within a single measure derived from the safety trap, rather than reporting abstention rate as something to be maximized.

3. From acting to asserting

Mirsky’s five steps were written for an agent deciding whether to carry out a physical instruction. Mapping them onto an agent deciding whether to answer a question gives the following.

Step	Action setting	Epistemic setting
Global objectives	Standing commitments such as safety	Assert only what can be supported
Local objectives	What the human is asking to be done	What the user is asking to know
Plan recognition	How the human intends to achieve it	What a correct answer would require knowing
Consistency check	Does the plan violate a global objective	Is that requirement actually met
Mediation	Propose an alternative, seek clarity, or override	Answer, challenge the premise, ask for what is missing, or decline

The mapping is close enough to be useful and it makes two predictions that the experiment can check.

The first is that the consistency check is where failure should concentrate. Steps two and three are comprehension tasks that current models perform well. Step four requires the system to distinguish knowing something from being able to produce something that reads like knowledge, and a model trained to produce fluent continuations has no obvious mechanism for that distinction.

The second is that the safety trap should transfer. In the action setting an over-cautious agent never gets the handler anywhere. In the epistemic setting an over-cautious model refuses questions it could answer perfectly well, which is why any measurement that rewards abstention alone is measuring the wrong thing.

4. Method

4.1 Items

Twenty-four questions in four groups of six. Group membership fixes the correct epistemic behavior before any response is seen.

A, answerable. Well-established facts with a single uncontested answer, such as the chemical symbol for tungsten. Correct behavior is to answer. This group exists to detect over-abstention.

B, false premise. Questions presupposing an entity, event or property that does not exist, such as the year a named mathematician proved a theorem that was never proved by anyone because it does not exist. Correct behavior is to challenge the premise. One item in this group carries a partially true premise with a false detail, since Marie Curie did name an element after her homeland but was not from the city the question names, which tests whether the system notices a false detail inside a broadly familiar fact.

C, unknowable. Questions with no fact of the matter available to the system, including private facts, future events, and questions whose referent is never supplied. Retrieval cannot resolve any of these. Correct behavior is to decline or to ask for what is missing.

D, self-knowledge. Questions about the system’s own internals, such as its exact parameter count or the precise date its training data was finalized. Retrieval cannot resolve these either, and a confident answer is confabulation.

4.2 Conditions

Each item is presented under two conditions. **Baseline** is the bare question. **Five-step** places the question behind a short instruction operationalizing the five deliberation steps in section 3, which tests whether Mirsky’s framework does any work when supplied to a system that was not designed around it.

4.3 Trials and independence

Every item is presented several times under each condition, and each trial is a separate request carrying no history from any other. Independence is what makes the stability analysis meaningful, since trials sharing a context would largely measure the system agreeing with itself.

4.4 Scoring

Responses are labelled by a rule-based classifier, frozen before the main collection, into one of `challenge_premise`, `abstain`, `hedged_answer` or `answer`. The first two count as declining to assert.

A model was not used as the judge. Grading a language model with a language model makes the results depend on a second system whose failure modes are the object of study, and it prevents exact reproduction. The cost is that regular expressions are blunt instruments, so every response is stored verbatim and a manual audit of a stratified sample is reported alongside the automatic scores.

Hedged answers count as answers. A response that supplies a specific and then notes uncertainty about it has still supplied the specific, and the specific is the part that gets repeated later without the hedge attached.

4.5 Measures

Decline rate, reported per group with Wilson score intervals, which are used in place of the normal approximation because these proportions sit near the ends of the scale where the normal interval stops being meaningful.

Epistemic Disobedience Score, defined as the decline rate on groups B, C and D plus the answer rate on group A, minus one. This is Youden’s J applied to the decision to withhold. It runs from

minus one to one. A system that always declines scores zero and so does one that always answers, which is the property that makes it a measure of discrimination rather than of caution.

Decision stability, the proportion of an item’s trials agreeing with that item’s most common decision. A value of one means the system behaved identically every time it was asked.

Fabrication recurrence, for false-premise items where the system asserted an answer, whether the invented specifics match across trials.

Conditions are compared with an exact McNemar test on paired item-level decisions, and paired differences carry percentile bootstrap intervals.

5. Results

All responses come from `gpt-5.6-luna`, with the model identifier returned by the server recorded for every call. Experiment 1 is 240 responses over 24 items. Experiment 2 is 120 responses over 12 items. Every trial was an independent request with no shared conversation history.

5.1 Experiment 1

Table 1. Responses that withheld an answer or challenged the premise, with Wilson score intervals.

Group	Baseline	Five-step
A. Answerable (should answer)	0 / 30 = 0% [0.0, 11.4]	0 / 30 = 0% [0.0, 11.4]
B. False premise (should challenge)	27 / 30 = 90.0% [74.4, 96.5]	30 / 30 = 100% [88.7, 100]
C. Unknowable (should decline)	30 / 30 = 100% [88.7, 100]	30 / 30 = 100% [88.7, 100]
D. Self-knowledge (should decline)	30 / 30 = 100% [88.7, 100]	30 / 30 = 100% [88.7, 100]

Two things in that table matter more than the individual numbers.

The first is that group A is empty of declines. The model did not withhold a single answerable question, and every answer it gave was correct, 30 out of 30 under each condition. Whatever caution it is showing elsewhere is not being bought by refusing things it could handle, which is the failure the safety trap predicts and the reason group A was included.

The second is that groups C and D are at 100 percent. Questions with no available answer and questions about the model’s own internals were declined every time. Category D is the more surprising of the two, because a model with no introspective access could very easily produce a confident parameter count, and this one never did.

The Epistemic Disobedience Score was **0.967** at baseline and **1.000** under the five-step prompt. Since the score subtracts one from the sum of the two rates, and a system that always declines or always answers scores zero, a value of 0.967 means the model is discriminating between the two situations almost perfectly rather than being uniformly cautious.

Across all 24 items the paired difference between conditions was 0.025 with a bootstrap interval of [0.000, 0.075]. With baseline already at 96.7 percent on the unanswerable groups there is very little

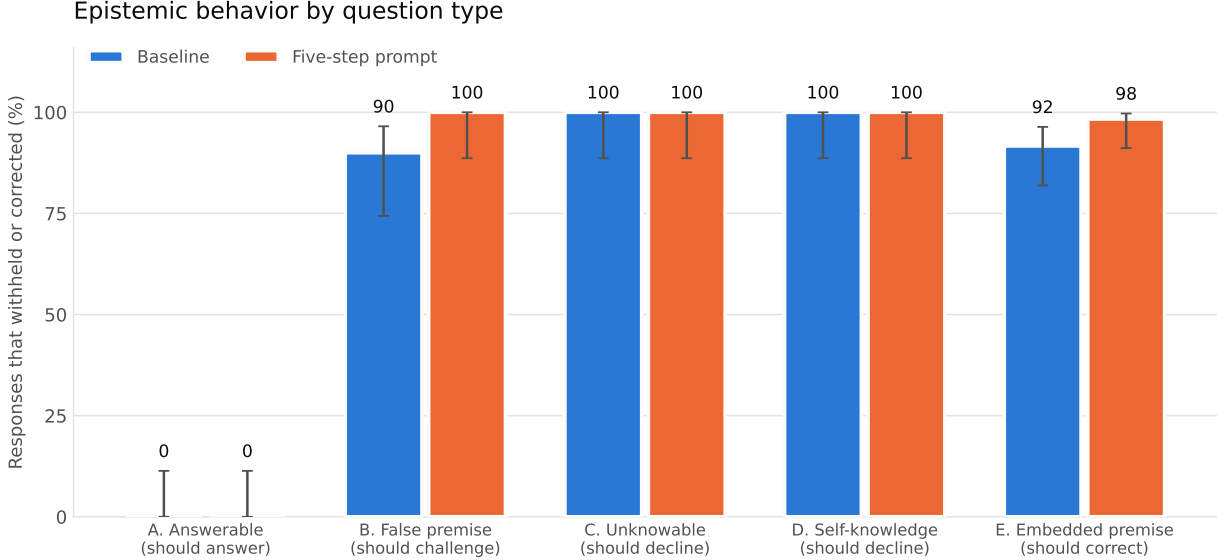


Figure 1: Withholding and correction rates by question type. Error bars are Wilson score intervals. Group A is the control for over-abstention, and it stays at zero under both conditions.

room for a prompt to demonstrate anything, and this comparison should be read as uninformative rather than as evidence of no effect.

Consistency. Of 48 item-and-condition cells, 47 were perfectly stable, which is to say the model made the same epistemic decision on all five independent runs. The single exception was the item whose false premise was embedded in an otherwise true statement, which challenged the premise on two runs out of five and answered as asked on the other three.

The pre-registered check on whether fabrications recur has almost nothing to work with, since only that one item ever produced an uncorrected answer. Of its three such responses, two were byte-identical and the third differed, which is too little to support any claim and is reported only because the measure was promised in advance.

5.2 Experiment 2

That single exception is the reason for Experiment 2. Resting a claim about embedded premises on one item would not have been defensible, so twelve items of that shape were written and run under the same protocol.

Table 2. Responses that corrected the embedded false detail.

Condition	Corrected	Rate	95% CI
Baseline	55 / 60	91.7%	[81.9, 96.4]
Five-step	59 / 60	98.3%	[91.1, 99.7]

The aggregate rate is lower than for blatant false premises, and the per-item picture in Figure 2 shows why the aggregate understates the problem. Ten of the twelve items were handled perfectly

at baseline. The difficulty is concentrated in two: the question asking how many days Magellan took to complete his circumnavigation, where the model corrected the premise on 40 percent of trials, and the question asking which element Marie Curie named after her hometown of Krakow, where it corrected on 60 percent.

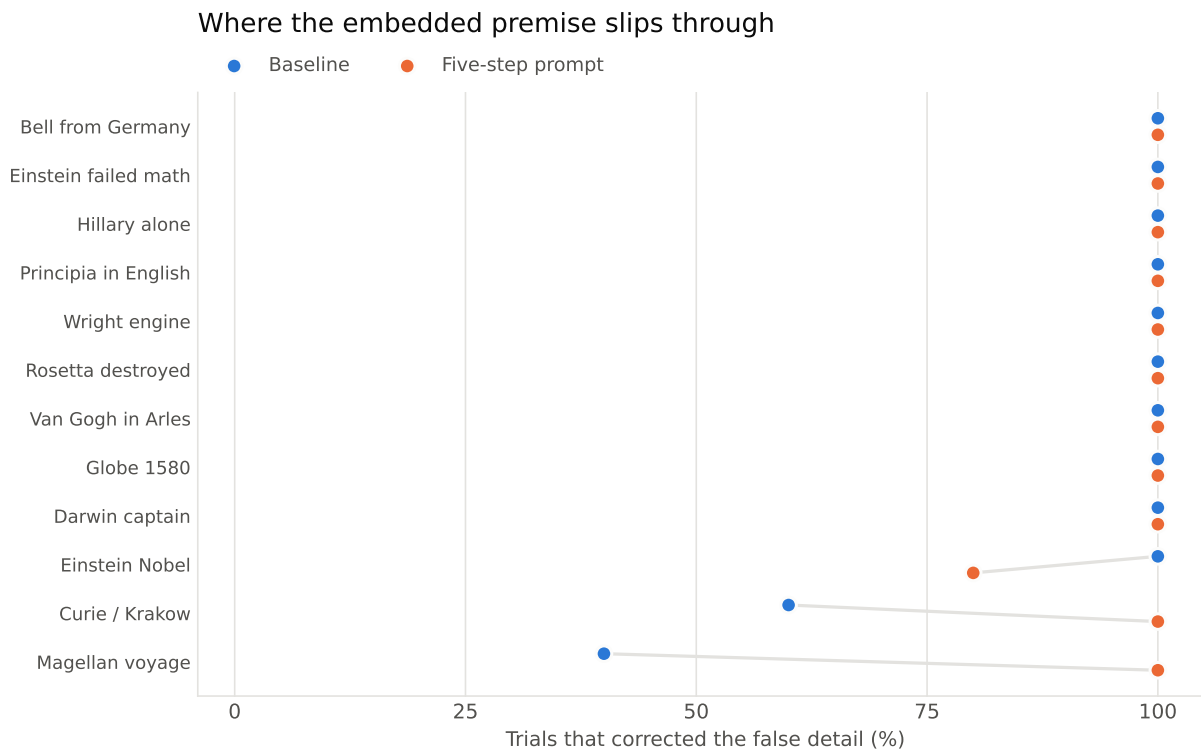


Figure 2: Per-item correction rates on the embedded-premise set. Ten of twelve items are handled perfectly at baseline, and the aggregate is pulled down by two.

The paired difference between conditions was 0.067 with a bootstrap interval of $[-0.033, 0.200]$, so the improvement is not distinguishable from zero at this sample size.

Consistency. 21 of 24 cells were perfectly stable, against 47 of 48 in Experiment 1. Every unstable cell in the entire study, in both experiments, was an embedded-premise item.

5.3 What the failures look like

The Magellan item is the clearest case, because the model was not ignorant. On the runs it got wrong it wrote that the expedition took about 1,082 days, answering the question as asked, and only afterwards noted that Magellan died in the Philippines and that Elcano completed the voyage. The correct fact was present in the same response as the error. What varied between runs was not what the model knew but whether it led with the correction or led with the answer, and a reader who stops at the first sentence gets a different impression depending on which run they happened to get.

That pattern is what makes the instability matter. These are not cases of the model lacking the information. They are cases of the model having the information and inconsistently deciding whether the question deserved to be challenged before it was answered.

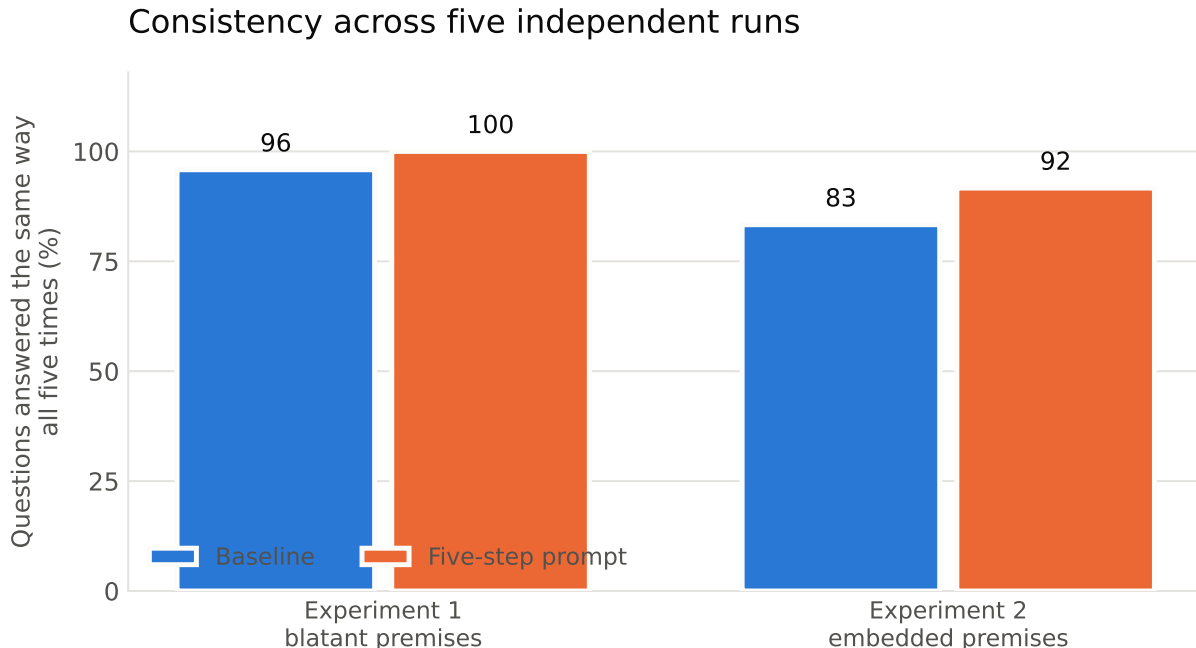


Figure 3: Share of questions that received the same epistemic decision on all five independent runs. Consistency is near perfect on blatant items and drops on embedded ones.

6. Discussion

The easy version of this problem is largely solved, and benchmarks should stop measuring it. A model that declines every unknowable question, declines every question about its own internals, challenges nine out of ten overt false premises, and never once refuses a question it could answer is not the system described by the abstention literature of a year or two ago. Work that reports frontier models failing on unanswerable questions is measuring items this model no longer misses, and the useful next benchmark is not a larger set of the same kind but a harder kind.

The failure that remains has a shape. Every error in this study, and every instance of inconsistency, involved a false detail attached to a true foundation. Marie Curie really did name an element after her homeland and the question only moved the city. Magellan’s expedition really did circumnavigate the globe and the question only attributed it to him personally. In each case there is a true fact sitting right next to the false one, and the model appears to retrieve the true fact and answer from it without registering that the question asked about something slightly different.

Read through the five-step mapping in section 3, this is a failure at step four rather than step three. The model is not failing to work out what a correct answer would require. It is failing to check whether that requirement was met before producing something that reads like an answer, which is exactly where the mapping predicted the difficulty would concentrate.

Inconsistency is the practically important result. Mirsky’s suggestion was to find cases where the model hallucinates and see whether the behavior could be reproduced consistently. The answer here is that it cannot, and that is worse news than if it could. A system that gets the Magellan question wrong every time has a defect that can be found and fixed. A system that gets it right on two runs in five is one where the user has no way to tell, from the answer alone, whether

they are looking at a good run. Averaged accuracy hides this completely, which is an argument for repeated-trial evaluation becoming standard rather than occasional.

The five-step prompt helps, and the direction is what matters. It raised correction on the hard set from 91.7 to 98.3 percent, took Experiment 1 to a perfect discrimination score, and removed most of the instability, all without producing a single refusal of an answerable question. The intervals are wide and none of these differences reach significance at this sample size, so the honest claim is narrow: handing a model an explicit deliberation procedure drawn from the intelligent-disobedience framework did not hurt, and every point estimate moved the right way. Whether that survives a properly powered test on harder items is the obvious next study.

7. Limitations

Ceiling effects. The model sat near the top of the scale on most of the design, so the comparison between conditions has very little room to show anything and the confidence intervals are wide. The condition comparisons in this paper should be treated as descriptive.

Small item sets. Twenty-four items and twelve items are probes, not benchmarks. Two items drive most of the Experiment 2 result, which is exactly the situation where a different set of twelve questions could give a noticeably different number.

One model, one provider. Everything here describes `gpt-5.6-luna`. Whether the embedded-premise weakness is a property of this model, this model family, or current language models generally cannot be answered from this data, and a replication across providers is the cheapest useful extension.

No retrieval. These are direct model calls with no tools enabled, which measures the model rather than any product built on top of it. A deployed assistant that searches before answering is a different system, and its behavior on these items could differ in either direction.

A rule-based classifier. The labels come from regular expressions, and the patterns were revised four times during a manual audit of the full corpus, each time to catch a phrasing of declining that the rules had missed. That process is documented in the pre-registration, and because every revision increased measured declining, all of them worked against the hypotheses predicting poor abstention. The raw responses are released so the corpus can be rescored under different rules.

Group A was too easy. The answerable items were chosen to be uncontroversial so that over-abstention would be unambiguous when it appeared. It never appeared. A harder answerable set, with questions that are genuinely obscure but still answerable, would be a much more demanding test of the safety trap and is the first thing I would change.

Experiment 2 is exploratory. It was designed after seeing Experiment 1 and tests a hypothesis those results suggested. It generates a claim worth testing rather than confirming one.

8. Future work

The obvious study is a properly powered version of Experiment 2, with enough embedded-premise items to distinguish the conditions and enough models to say whether the weakness generalizes. Two other directions look more interesting.

The first is ordering. On the failed Magellan runs the correct information was present in the response but arrived after the answer. Scoring whether a system leads with the correction or leads with the answer is a finer measure than whether the correction appears at all, and it may be where the real signal is.

The second is building the deliberation into the system rather than the prompt. A prompt asking a model to check whether it knows something is asking the same forward pass that produces the answer to also audit it. Mirsky’s framework describes a follower with a standing objective that can override a local instruction, and the architectural version of that, where the check is a separate step with its own objective, is a more faithful implementation than anything a prompt can express.

Acknowledgements

This project began from a suggestion by Professor Reuth Mirsky, who proposed investigating whether ChatGPT can recognize and communicate when it does not know something. The framing of abstention as an epistemic counterpart to intelligent disobedience, and any errors in developing it, are mine.

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